P-256 trapdoor search, round 2 — run report

Run id: run-20261012T2150Z · deterministic probes · commit: see PR · source: RESEARCH_P256_BACKDOOR_SPECULATION.md §4 ranks 1–3

No	Hypothesis	Probe	Measured
1	H1 coefficients	ord(b) = ord(r) = ord(−3) mod p	253 bits, exact; contains the 160-bit prime Q of p−1
2	H1	ord(j) mod p	252 bits, exact; contains Q
3	H1	residuosity	b, r QR non-cubes; j QR cube; −b flips the QR bit
4	H1	Hamming weights	b z=+0.81, r z=−0.31, j z=−0.12, n z=+4.88 (n order-derived)
5	H1	perfect powers / nearest c·2^k	none; gaps ≥ 2^49 bits
6	H1	continued fractions (64 terms)	max partial quotients 1275, 62, 107
7	H2 prime	p−1 factorization	$2\cdot3\cdot5^2\cdot11\cdot101\cdot281\cdot1531\cdot65537\cdot490719\cdot6602113\cdot Q_{160}$ (complete)
8	H2	p+1 factorization	2^96·7·273921· 140-bit composite (partial)
9	H2	ord_p(2)	(p−1)/2 exactly (2 is a QR; p ≡ 7 mod 8)
10	H2	twist embedding degrees ≥2^20 subgroups	none for k ≤ 100

11	3 endo/isog	GLS gap $4p-t^2$	258 bits
12	H3	rational roots of x^3-3x+b	1 ($\Longrightarrow$ 1 rational 2-isogeny)

P-256 round-2 probes: recorded z-scores (run-20261012T2150Z, n=256 bits eac
±5σ anomaly threshold
Hamming-weight z-score vs binomial(256, 1/2)
4
2
0
−2
−4
b
r
j
n

Recorded values only; every number copied from records.json. No anomaly flags (threshold ±5σ).